

EME 165 Numerical Conduction Project

Assigned: Sunday May 18th 2025

Due: Sunday, June 8th 2025 by 11:59 pm

Firm deadline- no extensions

Points allocated: 15

1. This is an individual project. Derivations and coding should be the sole work of the individual submitting the work. Discussions with TA, faculty and peers in the class are permitted. Use of textbook is also permitted. Use of generative AI to debug the code is allowed. The student should clearly note how this resource was used in the submission. However, no online tutors (for example, tutors on sites like chegg.com) or other sources for any other part such as formulation of the project may be used.
2. You are required to use Matlab or a programming language to solve the problem. Use of *Excel, or a commercially available solver is not permitted.*
3. Begin your project as soon as it is assigned. Do not wait until the last minute. Formulation and debugging of the code can take longer than you may expect.
4. Submit a single pdf of derivation/formulation and the code and solution outputs in gradescope. In a separate gradescope assignment "Project 1 code" submit your code.

Pledge (Project will not be graded without a signed student pledge):

I confirm that my work in this project is **solely my own effort** and that I have not received assistance from **any person OR organization within or outside the class**. I understand that while discussion with friends is permitted, but the formulation and coding are my own work. I furthermore understand that any impropriety will be reported to the UC Davis student misconduct office for disciplinary action.

Student Signature: Avrim Madhira

If you are using blank sheets of paper for the solution rather than this printout, acknowledge that you have read the pledge above by signing your name such as:

Pledge: I have read and accept the pledge.

Signature: AM

Problem statement

Luke Skywalker is practicing with his lightsaber. The movement of the lightsaber in air generates a convective heat transfer coefficient of $h = 50 \text{ W/m}^2\text{-K}$. The freestream air is at $T_{\text{air}} = 20^\circ\text{C}$. The lightsaber consists of a laser core that is $L_1 = 120 \text{ cm}$ long and of diameter $R_{\text{core}} = 2 \text{ cm}$. The thermal conductivity of the core is $k_{\text{core}} = 0.35 \text{ W/m-K}$. The core generates volumetric heat non-uniformly as given by the expression below

$$\dot{q}'''(r, x) = \dot{q}'''(r = 0) \cdot \left(1 - \left(\frac{r}{R_{\text{core}}}\right)^2\right)$$

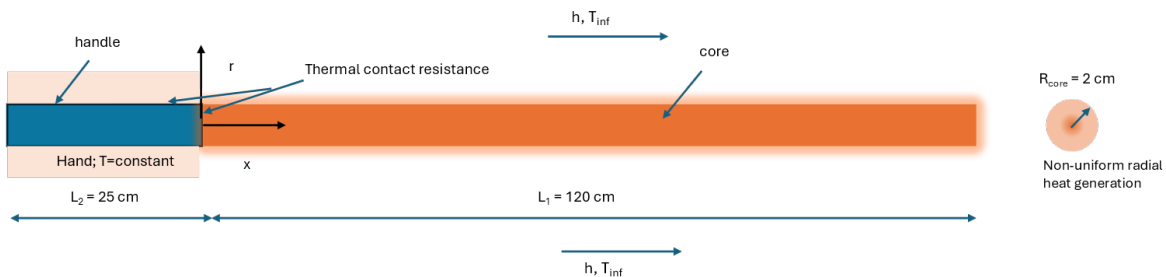
where $\dot{q}'''(r = 0) = 10^6 \text{ W/m}^3$ and $x=0$ is at the intersection of the laser core and the insulating handle.

The handle of length $L_2 = 25 \text{ cm}$ is made of a low thermal conductivity ceramic ($k_{\text{handle}} = 0.05 \text{ W/m-K}$). There is a contact resistance between the handle and the laser core of $R''_{\text{tc}} = 10^{-5} \text{ m}^2\text{-K/W}$. The figure shows the details including dimensions and boundary conditions.

Assume the entire handle of the sword is gripped firmly by Luke Skywalker and his hands are at a constant of 37.1°C and there is a thermal contact resistance of $R''_{\text{tc}} = 10^{-4} \text{ m}^2\text{-K/W}$ between the sword's handle and Luke's hand.



Credit: Star Wars, Luke Skywalker - Free Wallpaper / WallpaperJam.com



- (1) Develop a 2D steady state numerical conduction model for the lightsaber.

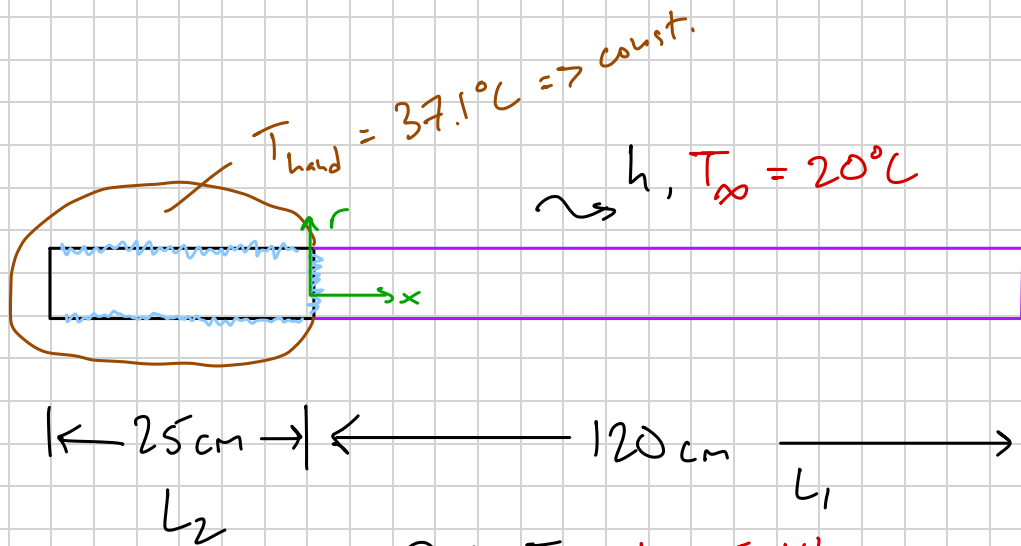
Identify the unique control volumes for which different energy balance equations need to be developed. Derive these equations for unique control volumes.

You need to have at least 5 nodes in the radial direction for the core and insulation regions. In the axial direction, use a minimum of 12 nodes in the axial direction for the core, and a minimum of 5 nodes in the axial direction for the handle.

- (2) Solve the system of equations to determine the temperature distribution within the lightsaber.
- (3) Plot a contour of the temperature distribution in the core and handle.
- (4) Determine the rate at which heat transfer is transferred to Luke's hand from the laser core.
- (5) Out of frustration, Luke plunges the lightsaber into a lake whose temperature is $T_{\text{lake}} = 10^\circ\text{C}$. The intense heat from the lightsaber causes the water surrounding the lightsaber to boil. Determine the new temperature distribution within the lightsaber. Assume a convective heat transfer coefficient of $5,000 \text{ W/m}^2\text{-K}$ during the boiling process.

Grading Rubric

Section	Points Allocated
1. Sketch identifying the origin, coordinate system, nodal numbering scheme, and unique control volumes for which energy balance equations need to be derived	1
2. Derivation of nodal equations for unique control volumes and pseudo code	8
3. Code completeness and accuracy	4
4. Contour plot of temperature (#3) with brief discussion of plot	1
5. Parts #4 and #5	1



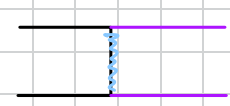
$\sim T_{\infty}, h = 50 \text{ W/m}^2\text{-K}$

$k_{\text{core}} = 0.35 \text{ W/m-K}$

$k_{\text{handle}} = 0.05 \text{ W/m-K}$

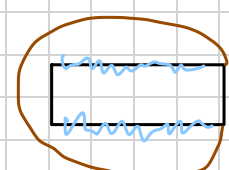
$\dot{q}'''(r, x) = \dot{q}'''(r=0) \cdot \left(1 - \left(\frac{r}{R_{\text{core}}}\right)^2\right)$

$R''_{tc} = 10^{-5} \frac{\text{m}^2\text{-K}}{\text{W}}$



- BC's: handle surface
- const. T BC
 - contact resistance $\rightarrow$ cond

$R''_{tc} = 10^{-4} \frac{\text{m}^2\text{-K}}{\text{W}}$



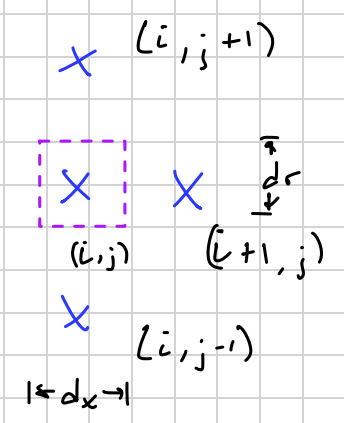
contact resistance handle $\leftrightarrow$ core

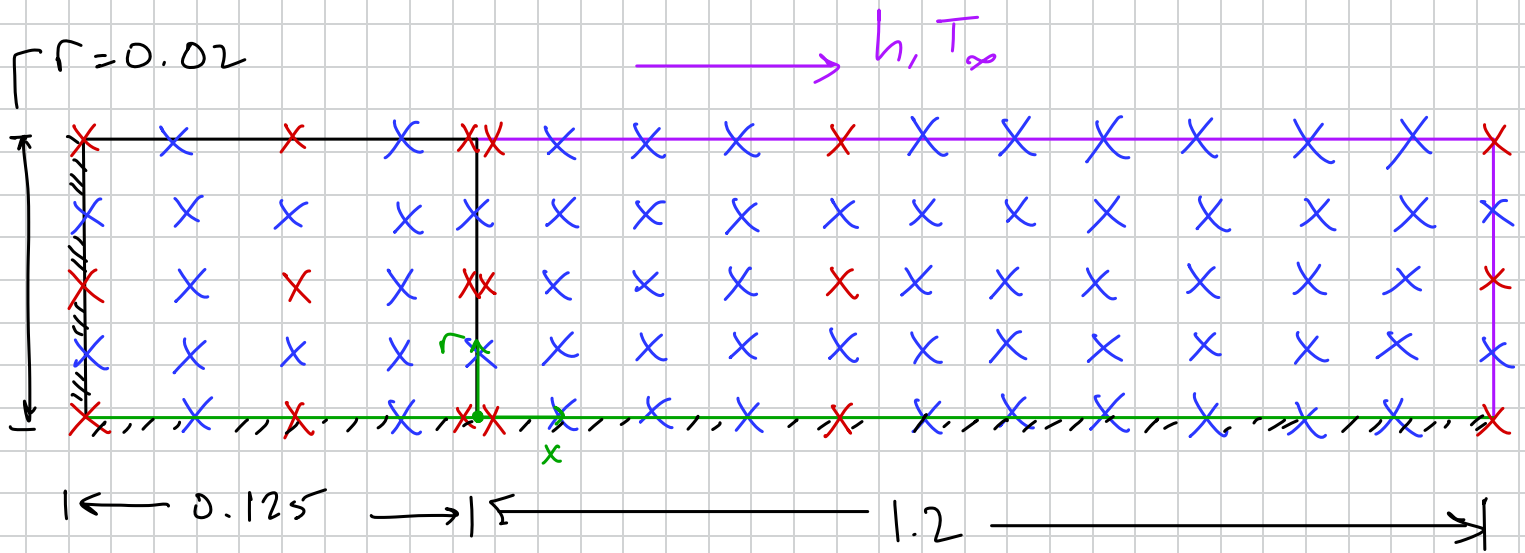
core

- convection to air

Code terms

$q_{x+} = kA \frac{(T_{i+1,j} - T_{i,j})}{dx} = q \times p(c/h)$
 $q_{x-} = kA \frac{(T_{i-1,j} - T_{i,j})}{dx} = q \times n(c/h)$
 $q_{r+} = kA \frac{(T_{i,j+1} - T_{i,j})}{dr} = q_{rp}$
 $q_{r-} = kA \frac{(T_{i,j-1} - T_{i,j})}{dr} = q_{rn}$

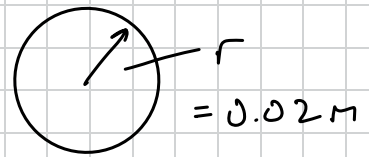




Internal Nodes:

$$dx_h = \frac{0.25}{N_{x_h} - 1}$$

$1 \rightarrow 5 = N_{x_h} - 1$



$$dr = \frac{0.02}{N_r - 1}$$

$1 \rightarrow 5 = N_r - 1$

$$dx_c = \frac{1.2}{N_{x_c} - 1}$$

$1 \rightarrow 12 = N_{x_c} - 1$

$$q_{(i-1,j \rightarrow i,j)} + q_{(i+1,j \rightarrow i,j)} + q_{(i,j-1 \rightarrow i,j)}$$

$$+ q_{(i,j+1 \rightarrow i,j)} + \cancel{q'' dx dr \cdot 1} = 0$$

$$\frac{k_c (2\pi r_j) (T_{i+1,j} - T_{i,j} + T_{i-1,j} - T_{i,j})}{dx}$$

①

handle

$$+ k_c \left[2\pi \left(r_j + \frac{dr}{2} \right) \right] dx \frac{(T_{i,j+1} - T_{i,j})}{dr} + k_c \left[2\pi \left(r_j - \frac{dr}{2} \right) \right] dx \frac{(T_{i,j-1} - T_{i,j})}{dr}$$

$$= 0$$

$$k_c (2\pi r_j) \frac{(T_{i+1,j} - T_{ij} + T_{i-1,j} - T_{ij})}{dx}$$

core

②

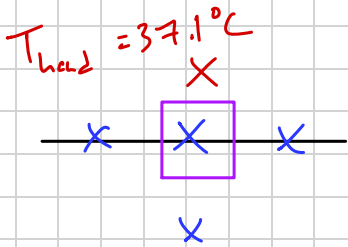
$$+ k_c \left[2\pi \left(r_j + \frac{dr}{2} \right) \right] dx \frac{(T_{i,j+1} - T_{ij})}{dr} + k_c \left[2\pi \left(r_j - \frac{dr}{2} \right) \right] dx \frac{(T_{i,j-1} - T_{ij})}{dr}$$

$$+ \dot{q}''' \cdot \pi \left[\left(r_j + \frac{dr}{2} \right)^2 - \left(r_j - \frac{dr}{2} \right)^2 \right] dx = 0$$

Handle

Boundary - cond handle $\rightarrow$ Luke's head

③



$$\dot{q}_{x-} + \dot{q}_{x+} + \dot{q}_{r-} + \dot{q}_{\text{contact}} = 0$$

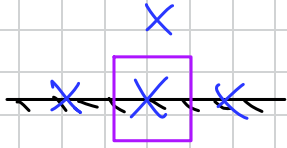
$$\dot{q}_{\text{contact}} = \frac{T_{ij} - T_{\text{head}}}{R_{tc}'' \cdot A_x}$$

$$k_h A_x \frac{(T_{i-1,j} - T_{ij})}{dx} + k_h A_x \frac{(T_{i+1,j} - T_{ij})}{dx} + k_h A_r \frac{(T_{i,j-1} - T_{ij})}{dr} + \frac{T_{ij} - T_{\text{head}}}{\underline{R_{tc}''} \cdot A_x} = 0$$

$$\star \left\{ \begin{array}{l} A_x = 2\pi r_j \\ A_{xc} = \pi r_j \\ A_r = 2\pi r_j dx \\ A_r = 2\pi \left(r_j \pm \frac{dr}{2} \right) dx \end{array} \right\} \star$$

$\rightarrow$ for $\dot{q}_{r-}$: (-) $\dot{q}_{r+}$: (+)

Boundary — handle → insulated surf (symmetry) (4)

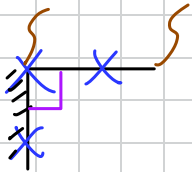


$$q_{x-} + q_{x+} + q_{r+} = 0$$

$$k_h A_x \frac{(T_{i-1,j} - T_{ij})}{dx} + k_h A_x \frac{(T_{i+1,j} - T_{ij})}{dx} + k_h A_r \frac{(T_{i,j+1} - T_{ij})}{dr} = 0$$

Corner — handle → insulated surf/hand

(5)

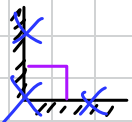


$$q_{x+} + q_{r-} + q_{\text{contact}} = 0$$

$$k_h A_{x_c} \frac{(T_{i+1,j} - T_{ij})}{dx} + k_h A_{r-} \frac{(T_{i,j-1} - T_{ij})}{dr} + \frac{(T_{ij} - T_{\text{hand}})}{A_{x_c} \cdot \underline{\underline{R''_{tc}}}} = 0$$

Corner — handle → insulated surface

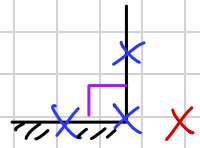
(6)



$$q_{x+} + q_{r+} = 0$$

$$k_h A_{x_c} \frac{(T_{i+1,j} - T_{ij})}{dx} + k_h A_{r+} \frac{(T_{i,j+1} - T_{ij})}{dr} = 0$$

Corner - handle $\rightarrow$ ins surf / core



$$q_{x-} + q_{r+} + q_{\text{contact}} = 0$$

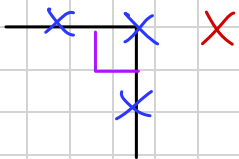
$$\frac{T_{ij\text{core}} - T_{ij}}{R_{\text{contact}}}$$

(7)

$$k_h A_{x_c} \frac{(T_{i-1,j} - T_{ij})}{dx} + k_h A_{r+} \frac{(T_{i,j+1} - T_{ij})}{dr} + \frac{T_{ij\text{core}} - T_{ij}}{A_{x_c} \cdot \underline{R_{tc}''}} = 0$$

Corner - handle $\rightarrow$ cir / core

(8)



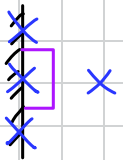
$$q_{x-} + q_{r-} + q_{\text{contact}} + q_{\text{conv}} = 0$$

$$k_h A_{x_c} \frac{(T_{i-1,j} - T_{ij})}{dx} + k_h A_{r-} \frac{(T_{i,j-1} - T_{ij})}{dr} + \frac{T_{ij\text{core}} - T_{ij}}{\frac{1}{2} A_{r-} \cdot \underline{R_{tc}''}} + h \left(\frac{1}{2} A_{r-} \right) (T_{\infty} - T_{ij})$$

= 0

Boundary - handle $\rightarrow$ ins surf

(9)

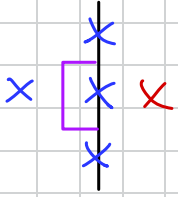


$$q_{x+} + q_{r-} + q_{r+} = 0$$

$$k_h A_x \frac{(T_{i+1,j} - T_{ij})}{dx} + k_h A_r \frac{(T_{i,j-1} - T_{ij})}{dr} + k_h A_{r+} \frac{(T_{i,j+1} - T_{ij})}{dr}$$

= 0

Boundary - ladle $\rightarrow$ core contact (10)

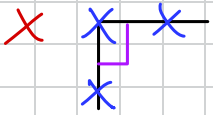


$$\dot{q}_{x-} + \dot{q}_{r+} + \dot{q}_{r-} + \dot{q}_{\text{contact}} = 0$$

$$k_h A_x \frac{(T_{i-1,j} - T_{ij})}{dx} + k_L A_{r+} \frac{(T_{i,j+1} - T_{ij})}{dr} + k_h A_{r-} \frac{(T_{i,j-1} - T_{ij})}{dr} + \frac{(T_{i,j_{\text{core}}} - T_{ij})}{A_{r+} \cdot \underline{\underline{R''_{tc}}}}$$

Core

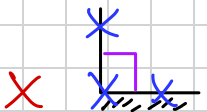
Corner - core $\rightarrow$ ladle/air



$$\dot{q}_{x+} + \dot{q}_{r-} + \dot{q}_{\text{contact}} + \dot{q}''' V = 0$$

$$k_c A_{x_c} \frac{(T_{i+1,j} - T_{ij})}{dx} + k_c A_{r-} \frac{(T_{i,j-1} - T_{ij})}{dr} + \frac{T_{i,j_L} - T_{ij}}{\frac{1}{2} A_{r-} \cdot \underline{\underline{R''_{tc}}}} + \dot{q}''' V = 0$$

Corner - core $\rightarrow$ ladle/ins surf (12)

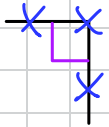


$$\dot{q}_{r+} + \dot{q}_{x+} + \dot{q}_{\text{contact}} + \dot{q}''' V = 0$$

$$k_c A_{r+} \frac{(T_{i,j+1} - T_{ij})}{dr} + k_c A_{x_c} \frac{(T_{i+1,j} - T_{ij})}{dx} + \frac{(T_{i,j_L} - T_{ij})}{\frac{1}{2} A_{r-} \cdot \underline{\underline{R''_{tc}}}} = 0$$

Corner - core $\rightarrow$ air

(13)

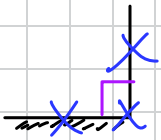


$$\dot{q}_{r-} + \dot{q}_{x-} + \dot{q}_{conv_x} + \dot{q}_{conv_r} + \dot{q}''' V = 0$$

$$k_c A_{r-} \frac{(T_{i,j-1} - T_{ij})}{dr} + k_c A_{x-} \frac{(T_{i-1,j} - T_{ij})}{dx} + h A_{x_c} (T_\infty - T_{ij}) + \frac{1}{2} h A_r (T_\infty - T_{ij}) + \dot{q}''' V = 0$$

Corner - core $\rightarrow$ air / ins surf

(14)

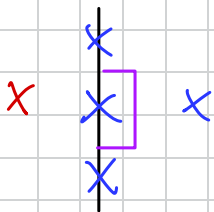


$$\dot{q}_{r+} + \dot{q}_{x-} + \dot{q}_{conv} + \dot{q}''' V = 0$$

$$k_c A_{r+} \frac{(T_{i,j+1} - T_{ij})}{dr} + k_c A_{x-} \frac{(T_{i-1,j} - T_{ij})}{dx} + \frac{1}{2} h A_r (T_\infty - T_{ij}) + \dot{q}''' V = 0$$

Boundary - core $\rightarrow$ handle contact

(15)

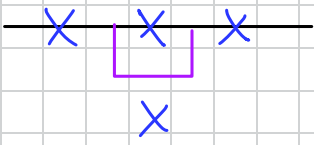


$$\dot{q}_{r+} + \dot{q}_{r-} + \dot{q}_{x+} + \dot{q}_{contact} + \dot{q}''' V = 0$$

$$k_c A_{r+} \frac{(T_{i,j+1} - T_{ij})}{dr} + k_c A_{r-} \frac{(T_{i,j-1} - T_{ij})}{dr} + k_c A_{x_c} \frac{(T_{i+1,j} - T_{ij})}{dx} + \frac{(T_{ijh} - T_{ij})}{A_{r-} \cdot \underline{\underline{R_{tc}''}}} + \dot{q}''' V = 0$$

Boundary - Core $\rightarrow$ air

(16)

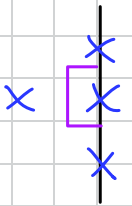


$$\dot{q}_{x-} + \dot{q}_{x+} + \dot{q}_{r-} + \dot{q}_{conv} + \dot{q}''' V = 0$$

$$k_c A_x \frac{(T_{i-1,j} - T_{ij})}{dx} + k_c A_x \frac{(T_{i+1,j} - T_{ij})}{dx} + k_c A_r \frac{(T_{i,j-1} - T_{ij})}{dr} + h A_x (T_\infty - T_{ij}) + \dot{q}''' V = 0$$

Boundary - Core $\rightarrow$ air

(17)

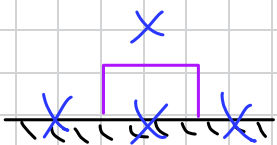


$$\dot{q}_{x-} + \dot{q}_{r+} + \dot{q}_{r-} + \dot{q}_{conv} + \dot{q}''' = 0$$

$$k_c A_x \frac{(T_{i-1,j} - T_{ij})}{dx} + k_c A_r \frac{(T_{i,j-1} - T_{ij})}{dr} + k_c A_r \frac{(T_{i,j+1} - T_{ij})}{dr} + \frac{1}{2} h A_r (T_\infty - T_{ij}) + \dot{q}''' V = 0$$

Boundary - Core $\rightarrow$ ins. surf

(18)



$$\dot{q}_{x-} + \dot{q}_{x+} + \dot{q}_{r+} + \dot{q}''' V = 0$$

$$k_c A_x \frac{(T_{i-1,j} - T_{ij})}{dx} + k_c A_x \frac{(T_{i+1,j} - T_{ij})}{dx} + k_c A_r \frac{(T_{i,j+1} - T_{ij})}{dr} + \dot{q}''' V = 0$$

$$\dot{q}''' V = \dot{q}''' \pi \left[\left(r_j + \frac{dr}{2} \right)^2 - \left(r_j - \frac{dr}{2} \right)^2 \right] dx \quad \star$$

Upper right corner core

$$T_{ij} =$$



$$\frac{k_c A_{r-} (T_{i,j-1})}{dr} + \frac{k_c A_{x_c} (T_{i-1,j})}{dx} + h A_{x_c} T_\infty + \frac{1}{2} h A_r T_\infty + \dot{q}''' V$$

$$\frac{k_c A_{r-}}{dr} + \frac{k_c A_{x_c}}{dx} + h A_{x_c} + \frac{1}{2} h A_r$$

Lower right corner core

$$T_{ij} =$$

$$\frac{k_c A_{r+} (T_{i,j+1})}{dr} + \frac{k_c A_{x_c} (T_{i-1,j})}{dx} + \frac{1}{2} h A_{r-} T_\infty + \dot{q}''' V$$

$$\frac{k_c A_{r+}}{dr} + \frac{k_c A_{x_c}}{dx} + \frac{1}{2} h A_{r-}$$

Upper left corner core

$$T_{ij} =$$

$$\frac{k_c A_{x_c} (T_{i+1,j})}{dx} + \frac{k_c A_{r-} (T_{i,j-1})}{dr} + \frac{T_{ij,h}}{\frac{1}{2} A_{r-} \cdot R_{tc}''} + \dot{q}''' V$$

$$\frac{k_c A_{x_c}}{dx} + \frac{k_c A_{r-}}{dr} + \frac{1}{\frac{1}{2} A_{r-} \cdot R_{tc}''}$$

Lower Left Corner Core

$$T_{ij} = \dot{q}''' V +$$

$$\frac{k_c A_{r+} (T_{i,j+1})}{dr} + \frac{k_c A_{xc} (T_{i+1,j})}{dx} + \frac{T_{ij,h}}{\frac{1}{2} A_{r-} \cdot R_{tc}''}$$

$$\frac{k_c A_{r+}}{dr} + \frac{k_c A_{xc}}{dx} + \frac{1}{\frac{1}{2} A_{r-} \cdot R_{tc}''}$$

Internal Core

$$T_{ij} = \dot{q}''' V +$$

$$\frac{k_c A_x (T_{i+1,j})}{dx} + \frac{k_c A_x (T_{i-1,j})}{dx} + \frac{k_c A_{r+} (T_{i,j+1})}{dr} + \frac{k_c A_{r-} (T_{i,j-1})}{dr}$$

$$\frac{k_c A_x}{dx} + \frac{k_c A_x}{dx} + \frac{k_c A_{r+}}{dr} + \frac{k_c A_{r-}}{dr}$$

Bottom edge Core

$$T_{ij} =$$

$$\frac{k_c A_x (T_{i-1,j})}{dx} + \frac{k_c A_x (T_{i+1,j})}{dx} + \frac{k_c A_{r+} (T_{i,j+1})}{dr} + \dot{q}''' V$$

$$\frac{k_c A_x}{dx} + \frac{k_c A_x}{dx} + \frac{k_c A_{r+}}{dr}$$

Top Edge Core

$$T_{ij} = h A_x T_\infty$$

$$\frac{k_c A_x (T_{i-1,j})}{dx} + \frac{k_c A_x (T_{i+1,j})}{dx} + \frac{k_c A_{r-} (T_{i,j-1})}{dr} + \dot{q}''' V$$

$$h A_x + \frac{k_c A_x}{dx} + \frac{k_c A_x}{dx} + \frac{k_c A_{r-}}{dr}$$

Core $\rightarrow$ handle contact

$$T_{ij} = \dot{q}''' V +$$

$$\frac{k_c A_{r+} (T_{i,j+1})}{dr} + \frac{k_c A_{r-} (T_{i,j-1})}{dr} + \frac{k_c A_x (T_{i+1,j})}{dx} + \frac{T_{ij} h}{A_{r-} \cdot R_{tc}''}$$

$$\frac{k_c A_{r+}}{dr} + \frac{k_c A_{r-}}{dr} + \frac{k_c A_x}{dx} + \frac{1}{A_{r-} \cdot R_{tc}''}$$

Upper Left Corner Handle

$$T_{ij} =$$

$$\frac{k_h A_{xc} (T_{i+1,j})}{dx} + \frac{k_h A_{r-} (T_{i,j-1})}{dr} + \frac{T_{hand}}{\frac{1}{2} A_{xc} \cdot R_{tc}''}$$

$$\frac{k_h A_{xc}}{dx} + \frac{k_h A_{r-}}{dr} + \frac{1}{\frac{1}{2} A_{xc} R_{tc}''}$$

Lower left corner handle

$$\frac{k_h A_{x_c} (T_{i+1,j})}{dx} + \frac{k_h A_{r+} (T_{i,j+1})}{dr}$$

$$= T_{ij}$$

$$\frac{k_h A_{x_c}}{dx} + \frac{k_h A_{r+}}{dr}$$

Upper Right corner handle

$$T_{ij} =$$

$$\frac{k_h A_{x_c} (T_{i-1,j})}{dx} + \frac{k_h A_{r-} (T_{i,j-1})}{dr} + \frac{T_{ij,cw}}{\frac{1}{2} A_{r-} \cdot R_{tc}''} + \frac{T_{head}}{A_{x_c} \cdot R_{tc}''}$$

$$\frac{k_h A_{x_c}}{dx} + \frac{k_h A_{r-}}{dr} + \frac{1}{\frac{1}{2} A_{r-} \cdot R_{tc}''} + \frac{1}{A_{x_c} \cdot R_{tc}''}$$

Lower Right corner handle

$$T_{ij} =$$

$$\frac{k_h A_{x_c} (T_{i-1,j})}{dx} + \frac{k_h A_{r+} (T_{i,j+1})}{dr} + \frac{T_{ij,cw}}{A_{x_c} \cdot R_{tc}''}$$

$$\frac{k_h A_{x_c}}{dx} + \frac{k_h A_{r+}}{dr} + \frac{1}{A_{x_c} \cdot R_{tc}''}$$

Internal Handle

$$T_{ij} =$$

$$\frac{k_c A_x (T_{i+1,j})}{dx} + \frac{k_c A_x (T_{i-1,j})}{dx} + \frac{k_c A_{r+} (T_{i,j+1})}{dr} + \frac{k_c A_{r-} (T_{i,j-1})}{dr}$$

$$\frac{k_c A_x}{dx} + \frac{k_c A_x}{dx} + \frac{k_c A_{r+}}{dr} + \frac{k_c A_{r-}}{dr}$$

Bottom Edge handle

$$T_{ij} =$$

$$\frac{k_L A_x (T_{i-1,j})}{dx} + \frac{k_L A_x (T_{i+1,j})}{dx} + \frac{k_L A_r (T_{i,j+1})}{dr}$$

$$\frac{k_L A_x}{dx} + \frac{k_L A_x}{dx} + \frac{k_L A_r}{dr}$$

Top Edge handle

$$T_{ij} =$$

$$\frac{k_L A_x (T_{i-1,j})}{dx} + \frac{k_L A_x (T_{i+1,j})}{dx} + \frac{k_L A_r (T_{i,j-1})}{dr} + \frac{T_{h,w2}}{A_x R_{tc}''}$$

$$\frac{k_L A_x}{dx} + \frac{k_L A_x}{dx} + \frac{k_L A_r}{dr} + \frac{1}{A_x \cdot R_{tc}''}$$

Handle $\rightarrow$ lukes hand

$$T_{ij} =$$

$$\frac{k_L A_x (T_{i+1,j})}{dx} + \frac{k_L A_r (T_{i,j-1})}{dx} + \frac{k_L A_{r+} (T_{i,j+1})}{dr}$$

$$\frac{k_L A_x}{dx} + \frac{k_L A_r}{dx} + \frac{k_L A_r}{dr}$$

Handle $\rightarrow$ core contact

$$T_{ij} =$$

$$\frac{k_L A_x (T_{i-1,j})}{dx} + \frac{k_L A_{r+} (T_{i,j+1})}{dr} + \frac{k_L A_{r-} (T_{i,j-1})}{dr} + \frac{T_{i,core}}{A_{r+} \cdot R_{tc}''}$$

$$\frac{k_L A_x}{dx} + \frac{k_L A_{r+}}{dr} + \frac{k_L A_{r-}}{dr} + \frac{1}{A_{r+} \cdot R_{tc}''}$$

Core $\rightarrow$ air

$$T_{cj} =$$

$$\frac{k_c A_x (T_{i-1,j})}{dx} + \frac{k_c A_{r-} (T_{i,j-1})}{dr} + \frac{k_c A_{r+} (T_{i,j+1})}{dr} + \frac{1}{2} h A_r T_\infty + \dot{q}''' V$$

$$\frac{k_c A_x}{dx} + \frac{k_c A_{r-}}{dr} + \frac{k_c A_{r+}}{dr} + \frac{1}{2} h A_r$$

0	252.4319	298.4890	317.9332	326.2444	329.7745	331.2688	331.9005	332.1674	332.2801	332.3277	332.3479	332.3564
95.0948	252.4359	298.4907	317.9339	326.2447	329.7747	331.2689	331.9005	332.1674	332.2801	332.3277	332.3479	332.3564
123.5781	251.1464	296.5821	315.9113	324.1752	327.6850	329.1705	329.7985	330.0638	330.1759	330.2233	330.2433	330.2517
137.4806	248.5881	293.1616	312.2903	320.4707	323.9444	325.4145	326.0359	326.2984	326.4094	326.4562	326.4760	326.4844
144.4055	244.7403	288.3118	307.1609	315.2230	318.6454	320.0937	320.7058	320.9644	321.0737	321.1198	321.1393	321.1476
146.9633	239.6120	282.0914	300.5872	308.4973	311.8539	313.2741	313.8743	314.1279	314.2350	314.2802	314.2993	314.3074
146.3829	233.2383	274.5624	292.6355	300.3611	303.6379	305.0239	305.6096	305.8571	305.9616	306.0058	306.0244	306.0323
143.3531	225.6760	265.7963	283.3810	290.8909	294.0741	295.4202	295.9890	296.2293	296.3308	296.3737	296.3918	296.3994
138.3064	217.0012	255.8759	272.9098	280.1739	283.2506	284.5512	285.1008	285.3329	285.4310	285.4724	285.4899	285.4973
131.5378	207.3074	244.8966	261.3200	268.3095	271.2674	272.5174	273.0455	273.2685	273.3628	273.4026	273.4194	273.4265
123.2612	196.7051	232.9666	248.7218	255.4097	258.2371	259.4315	259.9361	260.1492	260.2392	260.2772	260.2933	260.3001
113.6402	185.3216	220.2070	235.2382	241.5988	244.2849	245.4193	245.8984	246.1007	246.1862	246.2223	246.2376	246.2440
102.8045	173.3012	206.7519	221.0049	227.0142	229.5494	230.6191	231.0710	231.2619	231.3425	231.3766	231.3910	231.3970
90.8605	160.8056	192.7481	206.1700	211.8061	214.1807	215.1827	215.6058	215.7845	215.8929	215.8918	215.9053	215.9110
77.8974	148.0152	178.3557	190.8947	196.1375	198.3437	199.2744	199.6673	199.8332	199.9033	199.9329	199.9454	199.9507
63.9902	135.1304	163.7469	175.3529	180.1845	182.2154	183.0718	183.4333	183.5860	183.6505	183.6777	183.6892	183.6941
49.2018	122.3730	149.1067	159.7315	164.1363	165.9859	166.7656	167.0947	167.2337	167.2924	167.3172	167.3277	167.3321
33.5838	109.9881	134.6317	144.2300	148.1949	149.8584	150.5595	150.8554	150.9804	151.0332	151.0555	151.0649	151.0689
17.1750	98.2467	120.5305	129.0608	132.5756	134.0494	134.6705	134.9326	135.0433	135.0900	135.1098	135.1181	135.1216
0	87.4488	107.0222	114.4493	117.5066	118.7883	119.3284	119.5563	119.6526	119.6933	119.7104	119.7177	119.7200

[illegible][illegible][illegible]

[illegible][illegible][illegible][illegible]

119.7230	119.7230	119.7230	119.7230	119.7230	119.7230	119.7230	119.7230	119.7230	119.7230	119.7230	119.7230	119.7230	119.7229
----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Columns 105 through 117

332.3620	332.3612	332.3592	332.3546	332.3436	332.3176	332.2557	332.1090	331.7607	330.9343	328.9732	324.3209	313.2905
332.3620	332.3612	332.3592	332.3546	332.3436	332.3176	332.2557	332.1090	331.7607	330.9343	328.9732	324.3209	313.2905
330.2573	330.2565	330.2545	330.2499	330.2390	330.2131	330.1516	330.0058	329.6595	328.8378	326.8880	322.2625	311.2956
326.4899	326.4891	326.4872	326.4826	326.4718	326.4462	326.3854	326.2410	325.8983	325.0852	323.1557	318.5782	307.7251
321.1530	321.1522	321.1503	321.1458	321.1352	321.1099	321.0500	320.9078	320.5703	319.7693	317.8685	313.3592	302.6676
314.3128	314.3120	314.3101	314.3057	314.2953	314.2705	314.2118	314.0723	313.7414	312.9560	311.0923	306.6706	296.1865
306.0375	306.0367	306.0349	306.0306	306.0205	305.9963	305.9390	305.8029	305.4799	304.7135	302.8947	298.5794	288.3471
296.4045	296.4038	296.4020	296.3978	296.3880	296.3645	296.3088	296.1767	295.8631	295.1188	293.3525	289.1617	279.2236
285.5022	285.5015	285.4998	285.4958	285.4862	285.4636	285.4098	285.2821	284.9791	284.2600	282.5536	278.5045	268.9011
273.4312	273.4305	273.4289	273.4250	273.4158	273.3941	273.3424	273.2197	272.9286	272.2376	270.5978	266.7066	257.4764
260.3046	260.3039	260.3024	260.2987	260.2899	260.2691	260.2197	260.1025	259.8244	259.1642	257.5975	253.8795	245.0583
246.2483	246.2476	246.2462	246.2427	246.2343	246.2146	246.1677	246.0564	245.7923	245.1655	243.6778	240.1471	231.7684
231.4011	231.4005	231.3991	231.3958	231.3879	231.3693	231.3251	231.2201	230.9710	230.3798	228.9766	225.6462	217.7407
215.9147	215.9142	215.9129	215.9098	215.9024	215.8850	215.8436	215.7453	215.5121	214.9586	213.6448	210.5263	203.1218
199.9542	199.9537	199.9525	199.9496	199.9428	199.9266	199.8881	199.7969	199.5803	199.0663	197.8463	194.9501	188.0712
183.6973	183.6968	183.6957	183.6930	183.6868	183.6719	183.6365	183.5525	183.3533	182.8803	181.7578	179.0928	172.7612
167.3350	167.3346	167.3336	167.3312	167.3254	167.3119	167.2797	167.2032	167.0218	166.5913	165.5694	163.1431	157.3770
151.0715	151.0711	151.0702	151.0680	151.0629	151.0507	151.0217	150.9530	150.7899	150.4029	149.4841	147.3024	142.1166
135.1240	135.1236	135.1228	135.1209	135.1164	135.1056	135.0799	135.0190	134.8746	134.5317	133.7179	131.7854	127.1910
119.7228	119.7225	119.7218	119.7201	119.7161	119.7068	119.6845	119.6315	119.5059	119.2078	118.5001	116.8196	112.8241

Columns 118 through 130

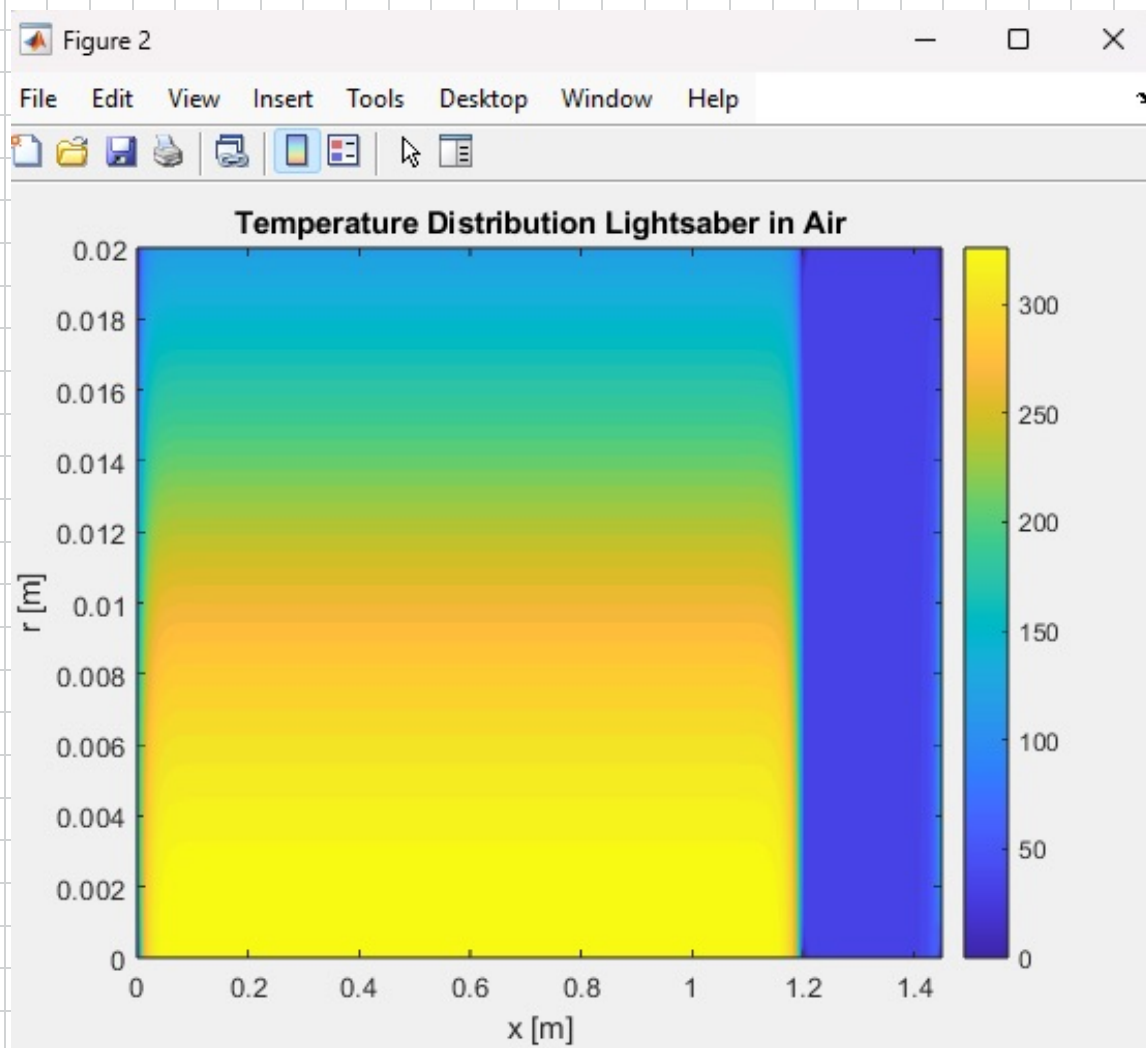
287.1430	224.5064	0	0	37.0888	37.0979	37.0995	37.0999	37.1000	37.1000	37.1000	37.1000	37.1001
287.1430	224.5064	48.5223	36.5799	37.0888	37.0979	37.0995	37.0999	37.1000	37.1000	37.1000	37.1000	37.1001
285.3034	223.2251	62.7365	37.0891	37.0909	37.0979	37.0995	37.0999	37.1000	37.1000	37.1000	37.1000	37.1001
282.0096	220.8095	69.5404	37.0997	37.0922	37.0980	37.0995	37.0999	37.1000	37.1000	37.1000	37.1000	37.1001
277.3424	217.2795	72.9078	37.1000	37.0930	37.0980	37.0995	37.0999	37.1000	37.1000	37.1000	37.1000	37.1001
271.3595	212.6612	74.2066	37.1000	37.0936	37.0981	37.0995	37.0999	37.1000	37.1000	37.1000	37.1000	37.1001
264.1212	206.9939	74.0734	37.1000	37.0940	37.0982	37.0995	37.0999	37.1000	37.1000	37.1000	37.1000	37.1000
255.6966	200.3300	72.8537	37.1000	37.0942	37.0983	37.0996	37.0999	37.1000	37.1000	37.1000	37.1000	37.1000
246.1653	192.7343	70.7525	37.1000	37.0944	37.0983	37.0996	37.0999	37.1000	37.1000	37.1000	37.1000	37.1000
235.6189	184.2838	67.8961	37.1000	37.0945	37.0984	37.0996	37.0999	37.1000	37.1000	37.1000	37.1000	37.1000
224.1606	175.0682	64.3620	37.1000	37.0946	37.0985	37.0997	37.0999	37.1000	37.1000	37.1000	37.1000	37.1000
211.9063	165.1898	60.1932	37.1000	37.0945	37.0986	37.0997	37.0999	37.1000	37.1000	37.1000	37.1000	37.1000
198.9844	154.7646	55.4052	37.1000	37.0945	37.0987	37.0997	37.0999	37.1000	37.1000	37.1000	37.1000	37.1000
185.5357	143.9234	49.9885	37.1000	37.0943	37.0989	37.0998	37.0999	37.1000	37.1000	37.1000	37.1000	37.1000
171.7138	132.8128	43.9084	37.1000	37.0941	37.0990	37.0998	37.1000	37.1000	37.1000	37.1000	37.1000	37.1000

Columns 131 through 140

37.1002	37.1010	37.1045	37.1192	37.1823	37.4531	38.6135	43.5632	63.7861	0
37.1002	37.1010	37.1045	37.1192	37.1823	37.4531	38.6136	43.5639	63.7888	95.0731
37.1002	37.1010	37.1044	37.1190	37.1815	37.4494	38.5977	43.4996	63.7451	123.5558
37.1002	37.1010	37.1043	37.1186	37.1799	37.4427	38.5690	43.3831	63.5190	137.4582
37.1002	37.1010	37.1042	37.1181	37.1777	37.4332	38.5285	43.2165	63.0650	144.3832
37.1002	37.1009	37.1041	37.1175	37.1749	37.4210	38.4767	43.0013	62.3647	146.9413
37.1002	37.1009	37.1039	37.1167	37.1715	37.4064	38.4143	42.7395	61.4148	146.3615
37.1002	37.1009	37.1037	37.1157	37.1675	37.3896	38.3420	42.4338	60.2219	143.3324
37.1002	37.1008	37.1034	37.1147	37.1631	37.3706	38.2607	42.0875	58.8001	138.2866
37.1002	37.1007	37.1032	37.1136	37.1582	37.3498	38.1714	41.7049	57.1695	131.5190
37.1002	37.1007	37.1029	37.1124	37.1530	37.3273	38.0752	41.2906	55.3555	123.2436
37.1001	37.1006	37.1026	37.1111	37.1475	37.3036	37.9732	40.8501	53.3879	113.6238
37.1001	37.1005	37.1023	37.1097	37.1417	37.2788	37.8666	40.3890	51.3010	102.7895
37.1001	37.1005	37.1019	37.1083	37.1357	37.2532	37.7567	39.9137	49.1330	90.8471
37.1001	37.1004	37.1016	37.1069	37.1296	37.2271	37.6449	39.4302	46.9265	77.8855
37.1001	37.1003	37.1013	37.1055	37.1235	37.2009	37.5324	38.9450	44.7281	63.9799
37.1001	37.1002	37.1009	37.1041	37.1174	37.1748	37.4204	38.4642	42.5887	49.1933
37.1000	37.1001	37.1006	37.1027	37.1115	37.1491	37.3103	37.9936	40.5640	33.5769
37.1000	37.1001	37.1003	37.1013	37.1056	37.1241	37.2033	37.5381	38.7147	17.1698
37.1000	37.1000	37.1000	37.1000	37.1000	37.1001	37.1005	37.1020	37.1069	0

Total heat transfer rate to Luke's hand: 0.20 W

>> |



[illegible][illegible][illegible]

223.6368	223.6368	223.6368	223.6368	223.6368	223.6368	223.6368	223.6368	223.6367	223.6366	223.6361	223.6345	223.6296	223.6147
223.6368	223.6368	223.6368	223.6368	223.6368	223.6368	223.6368	223.6368	223.6367	223.6366	223.6361	223.6345	223.6296	223.6147
221.5321	221.5321	221.5321	221.5321	221.5321	221.5321	221.5321	221.5321	221.5321	221.5319	221.5314	221.5298	221.5250	221.5059
217.7647	217.7647	217.7647	217.7647	217.7647	217.7647	217.7647	217.7647	217.7647	217.7645	217.7640	217.7625	217.7577	217.7420
212.4278	212.4278	212.4278	212.4278	212.4278	212.4278	212.4278	212.4278	212.4278	212.4276	212.4271	212.4256	212.4210	212.4068
205.5876	205.5876	205.5876	205.5876	205.5876	205.5876	205.5876	205.5876	205.5875	205.5874	205.5869	205.5855	205.5810	205.5606
197.3123	197.3123	197.3123	197.3123	197.3123	197.3123	197.3123	197.3123	197.3122	197.3121	197.3117	197.3103	197.3060	197.2962
187.6793	187.6793	187.6793	187.6793	187.6793	187.6793	187.6793	187.6793	187.6792	187.6791	187.6787	187.6774	187.6733	187.6600
176.7770	176.7770	176.7770	176.7770	176.7770	176.7770	176.7770	176.7770	176.7769	176.7768	176.7764	176.7752	176.7714	176.7559
164.7059	164.7059	164.7059	164.7059	164.7059	164.7059	164.7059	164.7059	164.7059	164.7058	164.7054	164.7043	164.7008	164.6859
151.5793	151.5793	151.5793	151.5793	151.5793	151.5793	151.5793	151.5793	151.5792	151.5791	151.5788	151.5778	151.5746	151.5685
137.5230	137.5230	137.5230	137.5230	137.5230	137.5230	137.5229	137.5229	137.5229	137.5228	137.5225	137.5216	137.5188	137.5059
122.6757	122.6757	122.6757	122.6757	122.6757	122.6757	122.6757	122.6757	122.6757	122.6756	122.6753	122.6745	122.6720	122.6649
107.1894	107.1894	107.1894	107.1894	107.1894	107.1894	107.1894	107.1894	107.1894	107.1893	107.1891	107.1884	107.1862	107.1775
91.2288	91.2288	91.2288	91.2288	91.2288	91.2288	91.2288	91.2288	91.2288	91.2287	91.2285	91.2280	91.2262	91.2200
74.9719	74.9719	74.9719	74.9719	74.9719	74.9719	74.9719	74.9719	74.9718	74.9718	74.9716	74.9712	74.9698	74.9655
58.6096	58.6096	58.6096	58.6096	58.6096	58.6096	58.6096	58.6096	58.6095	58.6095	58.6094	58.6091	58.6080	58.6040
42.3460	42.3460	42.3460	42.3460	42.3460	42.3460	42.3460	42.3460	42.3460	42.3460	42.3459	42.3457	42.3450	42.3422
26.3985	26.3985	26.3985	26.3985	26.3985	26.3985	26.3985	26.3985	26.3985	26.3984	26.3984	26.3983	26.3979	26.3939

Columns 113 through 126

223.5652	223.4117	222.9294	221.4136	216.6525	201.7179	155.0469	0	0	37.0888	37.0979	37.0995	37.0999	37.1000
223.5652	223.4117	222.9294	221.4136	216.6525	201.7179	155.0469	9.8811	36.5799	37.0888	37.0979	37.0995	37.0999	37.1000
221.4613	221.3094	220.8322	219.3325	214.6217	199.8438	153.6577	10.8170	37.0891	37.0909	37.0979	37.0995	37.0999	37.1000
217.6952	217.5463	217.0782	215.6072	210.9864	196.4885	151.1621	10.9403	37.0997	37.0922	37.0980	37.0995	37.0999	37.1000
212.3603	212.2154	211.7603	210.3298	205.8361	191.7339	147.6200	10.9393	37.1000	37.0930	37.0980	37.0995	37.0999	37.1000
205.5224	205.3829	204.9442	203.5656	199.2340	185.6372	143.0723	10.9122	37.1000	37.0936	37.0981	37.0995	37.0999	37.1000
197.2501	197.1169	196.6980	195.3817	191.2454	178.2573	137.5602	10.8751	37.1000	37.0940	37.0982	37.0995	37.0999	37.1000
187.6205	187.4946	187.0987	185.8544	181.9440	169.6606	131.1301	10.8310	37.1000	37.0942	37.0983	37.0996	37.0999	37.1000
176.7220	176.6043	176.2342	175.0708	171.4142	159.9233	123.8353	10.7809	37.1000	37.0944	37.0983	37.0996	37.0999	37.1000
164.6552	164.5464	164.2046	163.1301	159.7523	149.1325	115.7367	10.7253	37.1000	37.0945	37.0984	37.0996	37.0999	37.1000
151.5331	151.4340	151.1226	150.1438	147.0664	137.3865	106.9036	10.6646	37.1000	37.0946	37.0985	37.0997	37.0999	37.1000
137.4815	137.3927	137.1136	136.2363	133.4773	124.7951	97.4143	10.5994	37.1000	37.0945	37.0986	37.0997	37.0999	37.1000
122.6393	122.5612	122.3159	121.5445	119.1185	111.4804	87.3565	10.5302	37.1000	37.0945	37.0987	37.0997	37.0999	37.1000
107.1581	107.0911	106.8806	106.2185	104.1360	97.5764	76.8276	10.4573	37.1000	37.0943	37.0989	37.0998	37.0999	37.1000
91.2028	91.1471	90.9719	90.4213	88.6889	83.2298	65.9360	10.3795	37.1000	37.0941	37.0990	37.0998	37.1000	37.1000
74.9512	74.9068	74.7673	74.3289	72.9493	68.6001	54.8010	10.2884	37.1000	37.0939	37.0992	37.0998	37.1000	37.1000
58.5941	58.5611	58.4572	58.1304	57.1022	53.8597	43.5541	10.1404	37.0991	37.0936	37.0993	37.0999	37.1000	37.1000
42.3358	42.3139	42.2449	42.0282	41.3461	39.1944	32.3401	9.7227	37.0691	37.0933	37.0995	37.0999	37.1000	37.1000
26.3933	26.3822	26.3473	26.2376	25.8925	24.8035	21.3204	8.0042	36.0310	37.0930	37.0998	37.1000	37.1000	37.1000
10.9969	10.9962	10.9941	10.9874	10.9664	10.9002	10.6849	0	0	37.0994	37.1000	37.1000	37.1000	37.1000

Columns 127 through 140

37.1000	37.1000	37.1000	37.1000	37.1001	37.1006	37.1026	37.1114	37.1489	37.3105	38.0080	41.0323	53.8623	0
37.1000	37.1000	37.1000	37.1000	37.1001	37.1006	37.1026	37.1114	37.1489	37.3105	38.0081	41.0329	53.8649	73.2916
37.1000	37.1000	37.1000	37.1000	37.1001	37.1006	37.1026	37.1112	37.1484	37.3083	37.9984	40.9920	53.8403	95.0097
37.1000	37.1000	37.1000	37.1000	37.1001	37.1006	37.1026	37.1110	37.1474	37.3042	37.9809	40.9176	53.6832	105.3418
37.1000	37.1000	37.1000	37.1000	37.1001	37.1006	37.1025	37.1107	37.1461	37.2986	37.9562	40.8112	53.3602	110.1806
37.1000	37.1000	37.1000	37.1000	37.1001	37.1006	37.1024	37.1103	37.1444	37.2913	37.9246	40.6739	52.8596	111.5625
37.1000	37.1000	37.1000	37.1000	37.1001	37.1005	37.1023	37.1099	37.1424	37.2826	37.8866	40.5073	52.1819	110.4576
37.1000	37.1000	37.1000	37.1000	37.1001	37.1005	37.1022	37.1093	37.1401	37.2725	37.8427	40.3133	51.3355	107.4211
37.1000	37.1000	37.1000	37.1000	37.1001	37.1005	37.1020	37.1087	37.1375	37.2611	37.7934	40.0947	50.3350	102.8129
37.1000	37.1000	37.1000	37.1000	37.1001	37.1004	37.1019	37.1080	37.1346	37.2487	37.7393	39.8544	49.1999	96.8901
37.1000	37.1000	37.1000	37.1000	37.1001	37.1004	37.1017	37.1073	37.1315	37.2353	37.6812	39.5961	47.9537	89.8517
37.1000	37.1000	37.1000	37.1000	37.1001	37.1004	37.1015	37.1066	37.1282	37.2211	37.6198	39.3235	46.6234	81.8629
37.1000	37.1000	37.1000	37.1000	37.1001	37.1003	37.1013	37.1058	37.1247	37.2063	37.5558	39.0407	45.2393	73.0689
37.1000	37.1000	37.1000	37.1000	37.1001	37.1003	37.1011	37.1049	37.1212	37.1911	37.4900	38.7518	43.8343	63.6038
37.1000	37.1000	37.1000	37.1000	37.1001	37.1002	37.1010	37.1041	37.1176	37.1756	37.4232	38.4611	42.4440	53.5965
37.1000	37.1000	37.1000	37.1000	37.1000	37.1002	37.1008	37.1032	37.1140	37.1600	37.3562	38.1724	41.1061	43.1749
37.1000	37.1000	37.1000	37.1000	37.1000	37.1001	37.1006	37.1024	37.1103	37.1444	37.2897	37.8893	39.8606	32.4684
37.1000	37.1000	37.1000	37.1000	37.1000	37.1001	37.1004	37.1016	37.1068	37.1242	37.2245	37.6150	38.7491	21.6104

